Bonus 4: Row Averages vs the Average of Averages

Week 4 — guided double loop, Julia script submission

Submission: submit your Julia script on the course Moodle page.

Optional bonus: row averages vs the average of averages

Given any matrix A, compute the average of every row, then the average of those row averages, and count how many rows have an average strictly larger than the average of the row averages.

Task

Write a Julia program that works for any matrix A. You may assume that A is already defined when the code runs (the check provides it); do not define or read A yourself. The program:

- computes each row sum with a double loop and stores the row averages in a vector `row_mean`;
- computes the average of the row averages and stores it in `mean_of_means`;
- counts the rows with `row_mean[i] > mean_of_means` and stores the count in `count_above`.

Do not define functions; use only loops and `let` blocks. There is no need to print anything; the automatic check reads the three variables left by the program.

Submission

Submit your Julia script on the course Moodle page. The script must run from a fresh Julia process with A already defined by the check.

Checking

As a concrete check, take

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{pmatrix}.$$

Then the row averages are $2, 5, 8, 11$; their average is 6.5. Only the rows with averages 8 and 11 exceed 6.5, so the answer is 2. Your program must give exactly these values for this A, and still work when A is replaced by any other matrix.